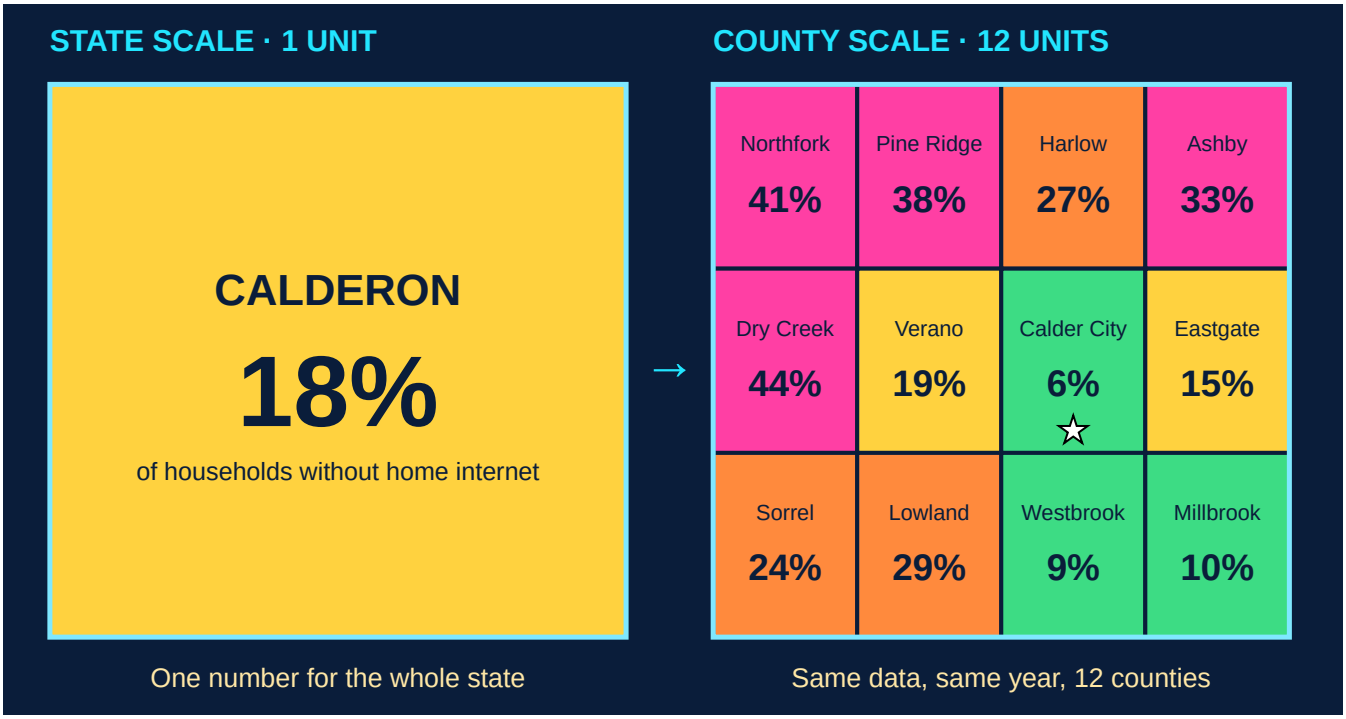
Two Scales, One State

AP Human Geography · Unit 1 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Both maps show the same variable for the same fictional state in the same year. The left panel reports one value for the whole state. The right panel divides Calderon into its 12 counties. The capital, Calder City, is marked with a star.

Percent of Households Without Home Internet, State of Calderon (fictional), 2020: State Scale vs. County Scale



0–10% 11–20% 21–30% 31–45% ★ State capital (Calder City)

Data and places are invented for practice. The state figure is a population-weighted average: most of Calderon's people live in the metro counties around Calder City, so those low rates pull the state number down. Shapes simplified, not to scale.

A. Identify the type of thematic map used in the county-scale panel of the stimulus. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names it: a choropleth map
- Describes the method: areas shaded or colored by data class

MODEL ANSWER

It is a choropleth map: each county is shaded one color based on which percent class it falls in.

Common mistake: "It is a political map of Calderon's counties." A political map shows borders and names, not data. This map colors each county by a percent, which makes it a thematic map with a specific name.

Why the model works: It names the map type precisely and adds a quick phrase showing it knows how the map works.

B. Using the map, describe how the pattern of households without home internet changes between the state scale and the county scale. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the state scale as one value or one shade (18%) for the whole state
- Describes the county-scale pattern with evidence: a wide range (6% to 44%), low near the capital or metro, high in rural or edge counties

MODEL ANSWER

T1- At the state scale there is no pattern to see, because the whole state is one shade with a single value of 18%.
E2- At the county scale the same data range from 6% in Calder City and 9% in Westbrook, clustered around the capital, up to 41% in Northfork and 44% in Dry Creek along the western and northern edges.
A3- Zooming in reveals a clustered pattern of rural counties where more than a third of households lack internet, which the single state number completely hides.

Common mistake: "At the county scale there are more colors and more details." That's about the picture, not the pattern. A describe answer needs the actual values (18% vs. 6% to 44%) and WHERE the high and low counties are.

Why the model works: It describes the state scale (one value, 18%), gives specific county evidence with places and numbers, and names what the zoom-in reveals (a rural cluster the state figure hides).

C. Explain ONE reason rural counties in Calderon have higher rates of households without home internet than the counties around the capital. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: low population density / homes spread far apart, high cost of infrastructure per customer, distance from the city, or lower incomes
- Explains the cause-and-effect chain to the higher rate

MODEL ANSWER

T1- Rural counties have higher rates of households without internet because their low population density makes internet infrastructure very expensive per customer.
E2- In Dry Creek (44%) homes are spread miles apart, so a company must lay far more fiber-optic cable to reach each household than it does in dense Calder City (6%), where one line serves hundreds of homes.
A3- Because the company earns less for every mile of cable in a rural county, it builds there last or not at all, so many rural households are left without service and the rate stays high.

Common mistake: “Rural counties don't have internet because they are in the country.” That restates the question: 'rural' and 'in the country' mean the same thing. You need the reason (homes far apart, costly cable) and the chain that leads to fewer connections.

Why the model works: It names the reason (low density, high cost per customer), proves it with map numbers (Dry Creek 44% vs. Calder City 6%), and explains the chain that leaves rural homes unconnected.

D. Using the map, explain ONE limitation of using only the state-level data to make decisions about improving internet access in Calderon. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- States the limitation: one state value is an average that hides or masks differences between counties
- Uses map evidence and explains the effect on a decision (wrong places, spread too thin, underestimating need)

MODEL ANSWER

T1- One limitation is that a single state-level number is an average that hides the differences inside the state, so planners can't see where the problem actually is.
E2- The map shows Calderon at 18%, but at the county scale Dry Creek is 44% and Northfork is 41% while Calder City is only 6%, and eight of the 12 counties are above the state figure.
A3- Because the 18% figure makes the whole state look moderately connected, a government using it alone might spread money evenly or underestimate the need, instead of targeting counties like Dry Creek and Northfork, where more than 40% of households lack internet.

Common mistake: “The state data is a limitation because it is from 2020 and might be out of date.” The question asks about using STATE-LEVEL data, so the limitation should be about scale, not the year. Compare 18% to the county values and explain what gets hidden.

Why the model works: It names what's hidden (county differences), proves it with map numbers (18% vs. 44% and 6%), and explains the decision that could go wrong.

E. Explain how a state government agency could use a geographic information system (GIS) to decide where to build new broadband internet infrastructure. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes what GIS does: layers, overlays, or combines different spatial data sets to analyze them
- Explains how that analysis leads to a location decision (where need is highest, where building is cheapest, which places to prioritize)

MODEL ANSWER

T1- A state agency could use GIS to layer several kinds of spatial data on top of each other and find the best places to build.

E2- For example, it could overlay this county map of households without internet with layers showing existing fiber-optic lines, population density, household income, and the locations of schools and clinics.

A3- Where the layers overlap, such as a rural county like Northfork with 41% unconnected, no nearby fiber, and a school, the agency can see where need is highest and building is possible, so it can target its limited money there first.

Common mistake: "The government could use GPS to find out where people live and then build internet there." That's GPS (finding locations), not GIS (layering and analyzing data). It also skips how the agency would decide WHICH places get built first.

Why the model works: It says what GIS does (layers data), names specific layers tied to the map, and explains how the overlap leads to choosing where to build.

F. Describe ONE way a dot density map or a graduated symbol map of the same data would show the pattern differently from the county choropleth. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the map type: dot density or graduated/proportional symbol
- Describes a real difference: it shows counts (number of households) instead of a rate, so metro counties could look bigger; or dots show where people cluster inside a county

MODEL ANSWER

T1- A dot density map would show the number of households without internet rather than the percent in each county.
E2- If each dot stood for 500 households and the dots were placed where people live, Calder City might have many dots even though its rate is only 6%, while Dry Creek at 44% might have only a few dots because so few people live there.
A3- This matters because it shows where the largest total number of unconnected households is, which can be a different place from where the rate is highest, and the dots can hint at clustering inside a county instead of coloring the whole county one shade.

Common mistake: "A dot density map would use dots instead of colors, so it would look different." That describes the drawing style, not the pattern. Say what the dots would SHOW: counts instead of percents, and where the dots would pile up.

Why the model works: It names the map type, describes exactly what would change (counts vs. rates, with Calder City and Dry Creek as proof), and says why that matters.

G. Explain how improved internet access in Calderon's rural counties is an example of time-space compression, and explain ONE effect it could have on a rural community. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Connects to time-space compression: distance matters less because places are connected faster or instantly
- Explains one effect on the rural community (remote jobs, telehealth, online school, online businesses, people staying rather than leaving)

MODEL ANSWER

T1- Better internet is an example of time-space compression because it shrinks the time it takes to connect a rural place with distant ones, so distance matters less.
E2- A family in Dry Creek that once drove two hours to see a specialist in Calder City could have a video visit in minutes, and a student there could take an online course from a university anywhere.
A3- One effect is that rural residents can reach jobs, health care, and education without leaving, which could help young people stay in the community and let local businesses sell to customers far away, slowing rural population loss.

Common mistake: “Time-space compression means the world is getting smaller because of technology.” That's a definition, not an application. You need to connect it to internet in Calderon's rural counties AND explain one effect on the community.

Why the model works: It ties internet access to the concept (shrinking the time to connect across distance), gives concrete examples from Dry Creek, and explains a community effect with a cause-and-effect chain.

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.